**Transport Phenomena ChE313 Final Exam** **12/9/2019**

92/100

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<u>Materials that you may use</u> during the exam:

1. One page (double sides) cheat sheet (A4). ONLY equations/formulas are allowed.
2. <u>NOT open book</u>
3. <u>NOT open notes</u> (including the printed handouts the instructors distributed in class)
4. <u>NOT open to any other resources</u> (including homework, quizzes, practical exercises).

** Violation of the above policy will result in significant grade deduction.*

**Short answer (3 pts each)**

3/3 (a) What is the overall tray efficiency in a distillation tower?

$$= \frac{\text{Number of Theor. Trays}}{\text{Number of Actual Trays}}$$

3/3 (b) What is the Morphree tray efficiency?

$$E_{ML} = \frac{X_n - X_{n-1}}{X_n^* - X_{n-1}}$$

$$E_{MV} = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}}$$

0/3 (c) What is the mass flux in SI units?

$$\frac{kmol}{s.m^2}$$

1/3 (d) What are the Fick's first and second laws of diffusion?

$$J = -D\, dc/dx$$

0/3 (e) What is the main assumption for MeCabe-Theile method of calculation for distillation?

**A. (10 pts)** Calculate the vapor and liquid compositions in equilibrium at 95 °C for benzene-toluene using the vapor pressure from the Table at 101.32 kPa.

*Vapor-Pressure and Equilibrium-Mole-Fraction Data for Benzene–Toluene System*

| Temperature | | Vapor Pressure | | | |
|---|---|---|---|---|---|
| | | Benzene | | Toluene | |
| K | °C | kPa | mm Hg | kPa | mm Hg |
| 353.2 | 80.1 | 101.32 | 760 | | |
| 358.2 | 85 | 116.9 | 877 | 46.0 | 345 |
| 363.2 | 90 | 135.5 | 1016 | 54.0 | 405 |
| 368.2 | 95 | 155.7 | 1168 | 63.3 | 475 |
| 373.2 | 100 | 179.2 | 1344 | 74.3 | 557 |
| 378.2 | 105 | 204.2 | 1532 | 86.0 | 645 |
| 383.8 | 110.6 | 240.0 | 1800 | 101.32 | 760 |

$P_A = 155.7$

$P_B = 63.3$

$x_A \cdot 155.7 + 63.3 \cdot (1 - x_A) = 101.32$

$x_A = 0.411$

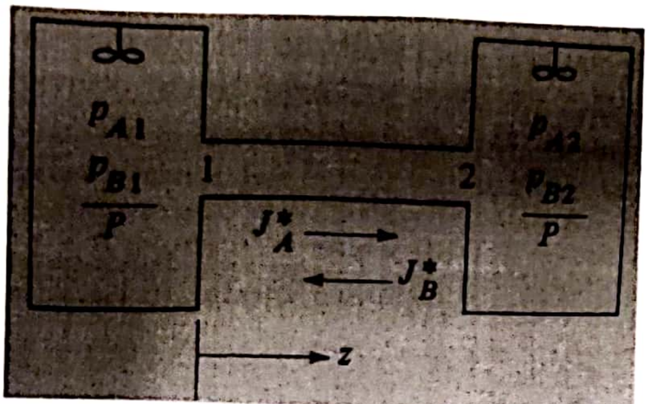
$x_B = 0.589$

$y_A = \frac{P_A x_A}{P} = 0.64$

**B. (10 pts)** Equimolar counter diffusion: Ammonia gas (A) is diffusing through a uiform tube 0.10 m long containing $N_2$ gas at $1.0132 \times 10^5$ Pa pressure and 298 K. At point 1, $p_{A1} = 1.013 \times 10^4$ Pa and at point 2, $p_{A2} = 0.507 \times 10^4$ Pa. The diffusivity $D_{AB} = 0.230 \times 10^{-4}$ m²/s.

(a) Calculate the flux $J^*_A$ at steady state.

(b) Calculate the flux $J^*_B$ at steady state.




(a) $J_A^* = \dfrac{(0.23 \times 10^{-4})(1.013 \times 10^4 - 0.5 \times 10^4)}{298(8314)(0.1)}$

$= 4.7 \times 10^{-7} \ \dfrac{\text{kg mol}}{\text{s.m}^2}$

(b) $J_B^* = \dfrac{(0.23 \times 10^{-4})(9.1 \times 10^4 - 9.6 \times 10^4)}{298(8314)(0.1 - 0)}$

$= -4.7 \times 10^{-7} \ \dfrac{\text{kgmol}}{\text{m}^2.\text{s}}$

**C. (15 pts)** The air in a room is at 26.7 °C and a apressure of 101.325 kPa and contains water vapor with a partial pressure $p_A = 2.76$ kPa. Assume air has 21% $O_2$ with a molecular weight of 32 and 79 % $N_2$ with a molecular weight of 28. The vapor pressure of water at101.325 kPa is 3.5 kPa. Calculate

(a) Humidity, H

(b) Saturation humidity, $H_s$, and percentage humidity, $H_p$

(c) Perentage relative humidity, $H_R$

(a)

$$H = \frac{18}{29} \cdot \frac{P_A}{P - P_A} = \frac{18}{29} \cdot \frac{2.76}{101.3 - 2.76} = \underline{0.0175}$$

(b)

$$H_s = \frac{18}{29} \cdot P_{AS}(P - P_{AS})^{-1} = 0.02$$

(c)

$$H_P = 100\% \times \frac{H}{H_s} = 78.3\%$$

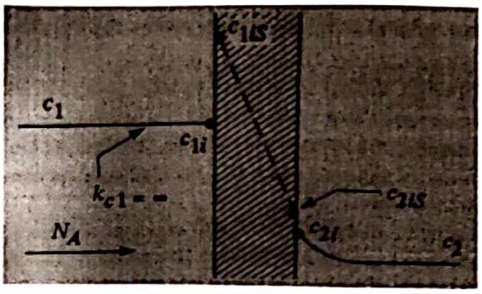
$$H_R = 100 \cdot \frac{P_A}{P_{AS}} = 78.9\%$$

20/20

**D. (20 pts)** A liquid containing dilute solute A at a concentration $c_1 = 3 \times 10^{-2}$ kmole/m³ is flowing rapidaly past a membrane of thickness $L = 3.0 \times 10^{-5}$ m. The distribution coefficient $K' = 1.5$ and $D_{AB} = 7.0 \times 10^{-11}$ m²/s in the membrane. The solute diffuses through the membrane, and its concentration on the other side is $c_2 = 0.5 \times 10^{-2}$ kmole/m³. The mass-transfer coefficient $k_{c1}$ is large and can be considered as infinite, and $k_{c2} = 2.0 \times 10^{-5}$ m/s.

(a) Derive the equation to calculate the steady state flux $N_A$.

(b) Calcuate the flux and the concentrations ($c_{1iS}$, $c_{2iS}$ and $c_{2i}$) at the membrane interfaces.




(a)

$$N_A = (c_1 - c_2) \Big/ \left(\frac{1}{K_{c1}} + \frac{1}{P_m} + \frac{1}{K_{c2}}\right) = \frac{c_1 - c_2}{\frac{1}{K_{c2}} + \frac{1}{P_m}}$$

steady state flux could be calculated with $\underline{K_{c1} = \infty}$

(b)

we need $P_m$:

$$P_m = \frac{K'D}{L} = \frac{7 \times 10^{-11} \times 1.5}{3 \times 10^{-5}} = 3.5 \times 10^{-6}$$

$$N_A = \frac{c_1 - c_2}{P_m^{-1} + K_{c2}^{-1}} = \underline{7.4 \times 10^{-8}}$$

$$N_A = K_{c2}(c_{2i} - c_2) \longrightarrow \underline{c_{2i} = 0.9 \times 10^{-2}}$$

$$K' = 1.5 = (c_{2is}) \cdot (c_{2i})^{-1} \longrightarrow \boxed{c_{2is} = 1.3 \times 10^{-2}}$$

30/30

**E. (30 pts)** A liquid mixture of benzene-toluene is to be distilled in a fractionating tower at 101.3 kPa pressure. The feed of 100 kmole/hr is saturated vapor, containing 45 mol% benzene and 55 mol% toluene, and enters at 366.7K. A distillate containing 95 mol% benzene and 5 mol% toluene and a bottom containing 10 mol% benzene and 90 mol% toluene are to be obtained. The reflux ratio is 4:1. The average heat capacity of the feed is 159 kJ/kmol.K and the average latent heat 32099 kJ/kmol. K. Calculate

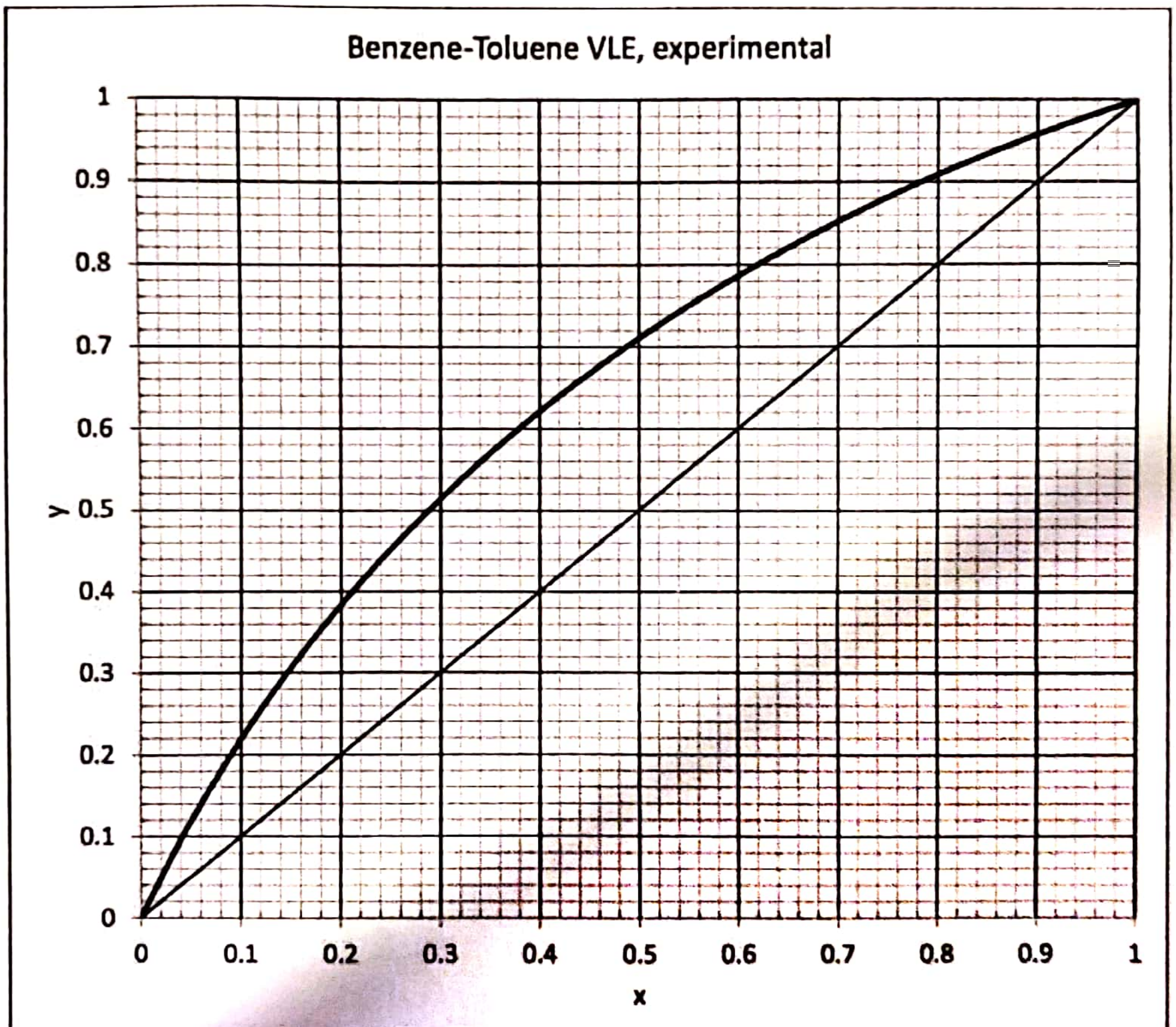
(a) the kmole per hour distillate

(b) kmole per hour bottoms

(c) q value

(d) the number of theoretical trays needed

(e) the minimal number of trays for the total reflux



mass Balance

$$\begin{cases} F = D + W \\ F x_f = D x_D + W x_W \end{cases}$$

$$\begin{cases} 100 = D + W \\ 100 \times 0.45 = D \times 0.95 + (100 - D)(0.1) \end{cases}$$

$$\begin{cases} D = 41.2 \text{ kgmol/h} \\ W = 58.8 \text{ kgmol/h} \end{cases}$$

$$\boxed{\begin{aligned} H_V - H_L &= 32099 \\ C_{P_L} &= 159 \end{aligned}}$$

$T_B = 366.7$

$$q = \frac{73800 + 38(200.3 - 130)}{13800} = 1.2$$

Slope of q line $= \frac{q}{q-1} = 6$

Enriching line:

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{X_D}{R+1} = 0.8 X_n + 0.19$$

5 Trays in Theory